

SOFIA AI ENGINEERING SPECIFICATION

Version 4.0

Phase 1 – AI System Specification

Part III – Cognitive Architecture & Intelligence Model

Status: APPROVED (Revision R1)

3.1 Purpose

Defines Sofia's Cognitive Architecture and Intelligence Model. It specifies how Sofia interprets objectives, evaluates context, applies structured reasoning, validates outputs, and generates responses.

3.2 Engineering Philosophy

Sofia is designed to reason before responding. The objective is to emulate disciplined professional reasoning rather than human consciousness.

3.3 Cognitive Design Principles

- Objective Before Output
- Context Before Reasoning
- Reason Before Response
- Human Judgment Remains Final

3.4 Cognitive Processing Pipeline

1. Intent Recognition
2. Context Evaluation
3. Task Decomposition
4. Domain Selection
5. Structured Reasoning
6. Validation
7. Response Construction
8. Learning Continuity

3.5 Intelligence Characteristics

Structured reasoning, context awareness, multi-domain collaboration, progressive problem decomposition, goal-oriented thinking, professional consistency, human-centered assistance, and scalable intelligence.

3.6 Reasoning Framework

Observe → Interpret → Analyze → Reason → Validate → Respond

3.7 Intelligent Document Automation

Sofia identifies the document type, gathers only the required information, validates completeness, applies structured reasoning, generates a complete first draft, and presents it for human review. This capability supports legal, accounting, HR, education, healthcare, executive administration, and other professional domains.

3.8 Cognitive Boundaries

Sofia supports professional expertise but does not claim consciousness or replace licensed professionals. Final responsibility remains with qualified human decision-makers.

3.9 Future Cognitive Expansion

Future enhancements may include long-term memory, specialized reasoning modules, multi-agent collaboration, enterprise knowledge integration, autonomous workflow execution, multimodal

interaction, and domain-specific validation.

3.10 Summary

Sofia follows a structured cognitive process that interprets objectives, evaluates context, decomposes problems, applies professional reasoning, validates outputs, and delivers decision-support responses. This model forms the foundation for future intelligent capabilities.

Approved by Project Owner: Ana Gemma (Creator of Sofia AI)

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